```
from google.colab import files
uploaded = files.upload()
```

Choose Files archive (2).zip
archive (2).zip(application/x-zip-compressed) - 9958 bytes, last modified: 3/11/2026 - 100% done
Saving archive (2).zip to archive (2).zip

```
import os
os.listdir()
```

```
['.config', 'archive (2).zip', 'sample_data']
```

```
import zipfile

with zipfile.ZipFile('archive (2).zip', 'r') as zip_ref:
    zip_ref.extractall()
```

```
import os
os.listdir()
```

```
['.config', 'loan_prediction.csv', 'archive (2).zip', 'sample_data']
```

```
import pandas as pd

data = pd.read_csv('loan_prediction.csv')
data.head()
```

	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	Coapplic
0	LP001002	Male	No	0	Graduate	No	5849	
1	LP001003	Male	Yes	1	Graduate	No	4583	
2	LP001005	Male	Yes	0	Graduate	Yes	3000	
3	LP001006	Male	Yes	0	Not Graduate	No	2583	
4	LP001008	Male	No	0	Graduate	No	6000	

Next steps: Generate code with data New interactive sheet

```
data.columns = data.columns.str.strip()
```

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 614 entries, 0 to 613
Data columns (total 13 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Loan_ID            614 non-null    object
 1   Gender             601 non-null    object
 2   Married            611 non-null    object
 3   Dependents         599 non-null    object
 4   Education          614 non-null    object
 5   Self_Employed      582 non-null    object
 6   ApplicantIncome    614 non-null    int64
 7   CoapplicantIncome  614 non-null    float64
 8   LoanAmount         592 non-null    float64
```

```
 9   Loan_Amount_Term   600 non-null    float64
 10  Credit_History     564 non-null    float64
 11  Property_Area      614 non-null    object
 12  Loan_Status        614 non-null    object
dtypes: float64(4), int64(1), object(8)
memory usage: 67.2+ KB
```

```
data.head()
```

	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	Coapplic
0	LP001002	Male	No	0	Graduate	No	5849	
1	LP001003	Male	Yes	1	Graduate	No	4583	
2	LP001005	Male	Yes	0	Graduate	Yes	3000	
3	LP001006	Male	Yes	0	Not Graduate	No	2583	
4	LP001008	Male	No	0	Graduate	No	6000	

Next steps: Generate code with data New interactive sheet

```
data.describe()
```

	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_History
count	614.000000	614.000000	592.000000	600.00000	564.000000
mean	5403.459283	1621.245798	146.412162	342.00000	0.842199
std	6109.041673	2926.248369	85.587325	65.12041	0.364878
min	150.000000	0.000000	9.000000	12.00000	0.000000
25%	2877.500000	0.000000	100.000000	360.00000	1.000000
50%	3812.500000	1188.500000	128.000000	360.00000	1.000000
75%	5795.000000	2297.250000	168.000000	360.00000	1.000000
max	81000.000000	41667.000000	700.000000	480.00000	1.000000

```
data.columns = data.columns.str.strip()
```

```
print(data.columns)
```

```
Index(['Loan_ID', 'Gender', 'Married', 'Dependents', 'Education',
       'Self_Employed', 'ApplicantIncome', 'CoapplicantIncome', 'LoanAmount',
       'Loan_Amount_Term', 'Credit_History', 'Property_Area', 'Loan_Status'],
      dtype='object')
```

```
data.isnull().sum()
```

	0
Loan_ID	0
Gender	13
Married	3
Dependents	15
Education	0
Self_Employed	32
ApplicantIncome	0
CoapplicantIncome	0
LoanAmount	22
Loan_Amount_Term	14
Credit_History	50
Property_Area	0
Loan_Status	0

dtype: int64

```
data = data.dropna()
```

```
print(data.columns)
```

```
Index(['Loan_ID', 'Gender', 'Married', 'Dependents', 'Education',
       'Self_Employed', 'ApplicantIncome', 'CoapplicantIncome', 'LoanAmount',
       'Loan_Amount_Term', 'Credit_History', 'Property_Area', 'Loan_Status'],
      dtype='object')
```

```
data.isnull().sum()
```

	0
Loan_ID	0
Gender	0
Married	0
Dependents	0
Education	0
Self_Employed	0
ApplicantIncome	0
CoapplicantIncome	0
LoanAmount	0
Loan_Amount_Term	0
Credit_History	0
Property_Area	0
Loan_Status	0

dtype: int64

```
data = data.fillna(method='ffill')
```

```
/tmp/ipykernel_226/917065080.py:1: FutureWarning: DataFrame.fillna with 'method' is deprecate
  data = data.fillna(method='ffill')
```

```
from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()

data['Gender'] = le.fit_transform(data['Gender'])
data['Married'] = le.fit_transform(data['Married'])
data['Dependents'] = le.fit_transform(data['Dependents'])
data['Education'] = le.fit_transform(data['Education'])
data['Self_Employed'] = le.fit_transform(data['Self_Employed'])
data['Property_Area'] = le.fit_transform(data['Property_Area'])
data['Loan_Status'] = le.fit_transform(data['Loan_Status'])
```

```
X = data.drop(['Loan_ID','Loan_Status'], axis=1)
y = data['Loan_Status']
```

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
!pip install imbalanced-learn
```

```
Requirement already satisfied: imbalanced-learn in /usr/local/lib/python3.12/dist-packages (0
Requirement already satisfied: numpy<3,>=1.25.2 in /usr/local/lib/python3.12/dist-packages (f
Requirement already satisfied: scipy<2,>=1.11.4 in /usr/local/lib/python3.12/dist-packages (f
Requirement already satisfied: scikit-learn<2,>=1.4.2 in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: sklearn-compat<0.2,>=0.1.5 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: joblib<2,>=1.2.0 in /usr/local/lib/python3.12/dist-packages (f
Requirement already satisfied: threadpoolctl<4,>=2.0.0 in /usr/local/lib/python3.12/dist-pack
```

```
from imblearn.over_sampling import SMOTE

smote = SMOTE()

X_train, y_train = smote.fit_resample(X_train, y_train)
```

```
from sklearn.linear_model import LogisticRegression

model = LogisticRegression()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWar
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:
    https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
    https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
  n_iter_i = _check_optimize_result(
```

```
from sklearn.metrics import classification_report, roc_auc_score

print(classification_report(y_test, y_pred))

print("ROC AUC:", roc_auc_score(y_test, y_pred))
```

```
              precision    recall  f1-score   support

           0       0.52      0.50      0.51        28
           1       0.80      0.81      0.80        68

    accuracy                           0.72        96
   macro avg       0.66      0.65      0.66        96
weighted avg       0.72      0.72      0.72        96

ROC AUC: 0.6544117647058824
```

```
from sklearn.tree import DecisionTreeClassifier

tree = DecisionTreeClassifier()

tree.fit(X_train, y_train)

y_pred_tree = tree.predict(X_test)

print(classification_report(y_test, y_pred_tree))
```

```
              precision    recall  f1-score   support

           0       0.71      0.54      0.61        28
           1       0.83      0.91      0.87        68

    accuracy                           0.80        96
   macro avg       0.77      0.72      0.74        96
weighted avg       0.79      0.80      0.79        96
```

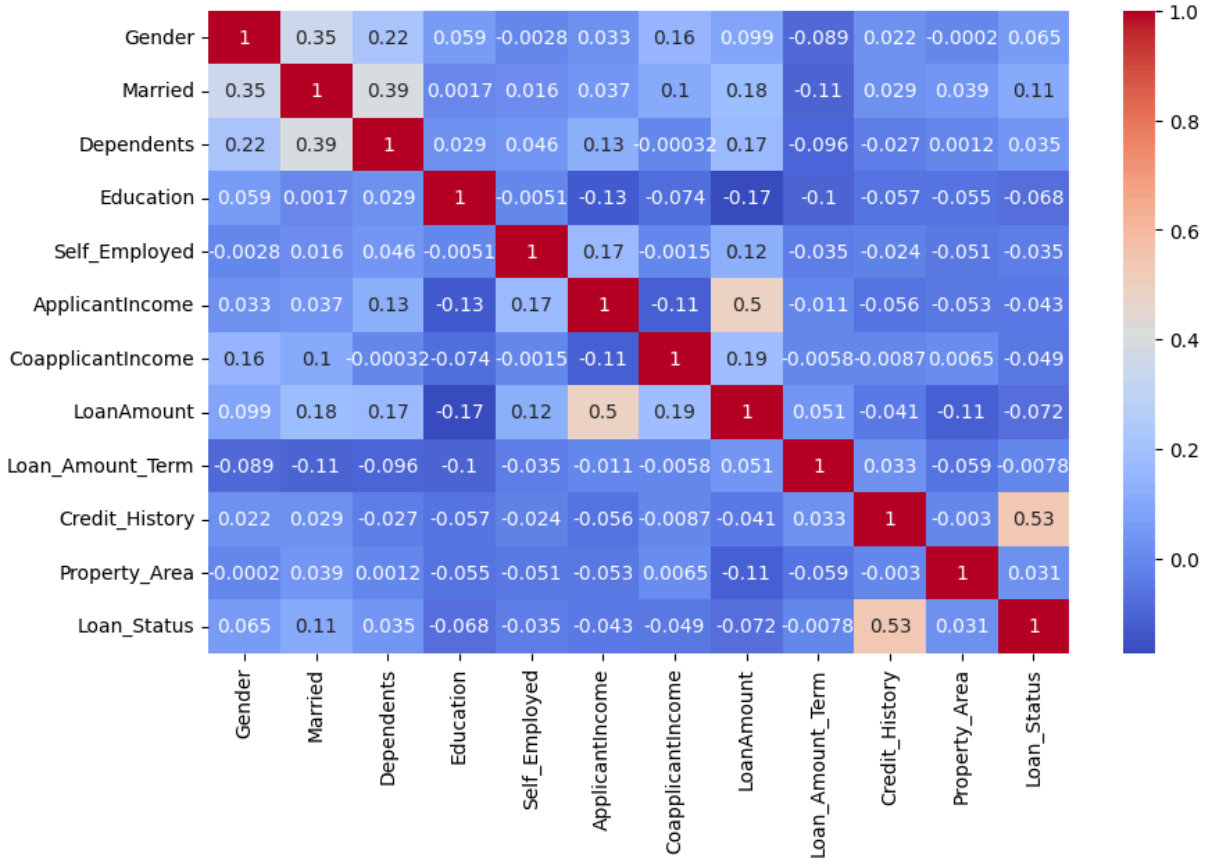
```
import seaborn as sns
import matplotlib.pyplot as plt

sns.countplot(x='Loan_Status', data=data)
plt.show()
```

```
plt.figure(figsize=(10,6))

sns.heatmap(data.drop('Loan_ID', axis=1).corr(), annot=True, cmap='coolwarm')

plt.show()
```



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